Day 18 - Pivot Tables: Monthly Trend and Top 5 Customers

Friday, 6 November 2026 | Swagat (Smooth) | Data Analytics sprint | File: pivot_practice_data.xlsx (32 rows)

1. Key ideas

- **Four areas:** Rows (categories listed down), Columns (categories listed across), Filters (limit the data shown), Values (the calculation).
- **Values can calculate:** Sum (default for numbers), Count, Average, Max, Min.
- **Dates group only if they are real dates** (stored as numbers), not text. Text months sort alphabetically: Aug, Nov, Oct, Sep. Fix used: a numeric MonthSort key (202608, 202609 ...) placed before Month in Rows forces the right order.
- **Chart choice:** line chart for values in a natural order (time), where slope shows direction. Bar chart for comparing separate categories (customers).
- **Top N:** Value Filters > Top 10 > Top 5 by Sum of Revenue (not Max). The filter picks the rows; sorting only changes display order.
- **Source range:** new rows are seen only after the range grows (Ctrl+T table, or Change Data Source) and you press Refresh.

2. Examples from the data

Monthly revenue	Top 5 customers
Aug 2026 = 120,500 Sep 2026 = 96,500 Oct 2026 = 81,500 Nov 2026 = 35,500 (partial: only Nov 1 and Nov 3) Grand Total = 334,000	Bina 89,500 | Dev 62,500 | Hari 54,000 Anil 41,000 | Esha 32,000 Top 5 sum = 279,000 Hidden by the filter: Chitra, Gita, Farid = 55,000 Check: 334,000 - 279,000 = 55,000

- Nov is not comparable with the other months: it is a partial month. Compare per-day revenue, or wait until the month ends.
- After a Top N filter, the pivot's Grand Total shows only the visible rows. It is not company revenue.

3. My mistakes and the fix

Mistake	Fix
Recall missed Count, Average, Max, Min, filter applies to the whole pivot, pivot chart, group dates, source range and Refresh.	List all five Values calculations. Remember: filters hit the whole pivot. New data needs the range updated and Refresh.
Recall said I grouped the date column, but I had deliberately used Month and MonthSort helper fields in Rows.	The method was valid. The mistake was in the description: name what I actually did. Alternative: group the real Order Date by Months (right-click > Group).
Line chart reason was circular: 'it shows comparison'.	Months have an order and the line connects them, so the slope shows increase or decrease. Comparison between categories is the bar chart's job.
Chart title was 'Total'; legend 'Total' carried no meaning.	Title it 'Monthly revenue'. Remove or rename a legend that adds nothing.
Said I sorted 'descending to ascending' before Top 5.	Sorting is not needed to get Top 5. Use the Top 10 value filter, By Sum of Revenue.
Did not state the Grand Total reason (279,000 vs 334,000).	State the cause and the numbers: filter hides 3 customers, 334,000 - 55,000 = 279,000.

4. How I improved

- Both numbers matched: all four monthly totals and the top 5 customers with their values.
- Later recall listed Sum, Max, Min, Count and Average, with nothing missing.
- Explained why November cannot be compared (partial month) and noticed the Grand Total was below total sales without being told.

5. Practice to retry (no solutions)

- Build a pivot of revenue by Region. State what goes in Rows and Values, choose a chart type, and give the reason in one sentence.
- Show the top 3 categories by revenue. Say what the Grand Total shows now, with the numbers.

- A Date column will not group by month. Give two possible causes and how to check each.
- Five new rows are added below the source data. What do you do so the pivot includes them?
- Show average revenue per order by month. Which Values setting do you change, and why is Sum the wrong answer here?

6. Recall questions

- Name the four pivot areas and one job of each, from memory.
- Why does a text month column sort wrongly, and what is the fix?
- After Top 5, why is the Grand Total not company revenue?
- Why is November not comparable, and what would you compare instead?

7. Open items

- Redo the hardest pivot problem from a blank sheet without notes.
- Optional: rebuild the monthly chart by grouping Order Date directly, and compare it with the MonthSort method.
- Read Day 18 recall again before the next weekly test.